

Aug 28, 2023

Introduction to Computer and Network Security

Part 1

Question 1 [40 points]

Define the notion of hash function and its main properties: (a) Ease of computation, (b) Compression, (c) One-way, (d) Weak collision resistance, and (e) Strong collision resistance.

Question 2 [10 points]

Explain the notion of pseudo anonymization function, describe at least two possible implementations together with their advantages and disadvantages.

Part 2

Question 3 [40 points]

Explain the notion of Multi-level security by illustrating it with the Bell-La Padula model; in particular discuss the notions of (a) security level, (b) need-to-know, (c) dominance relation, (d) no-read-up rule, (e) no-write-down rule.

Question 4 [10 points]

Briefly describe the notion of Data Protection Impact Assessment (DPIA) according to the Art. 35 of the GDPR.

Please, remember...

Be sure to annotate the following information on each used sheet of paper (possibly, 2 sheets at most)

- Name
- Surname
- Student ID number
- Question number